

The potential of the ensemble Kalman filter for NWP—a comparison with 4D-Var

By ANDREW C. LORENC*

Met Office, Exeter, UK

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SUMMARY

The ensemble Kalman filter (EnKF) is reviewed for its expected assimilation characteristics and ease of implementation, and compared to the currently more popular four-dimensional variational assimilation (4D-Var). The EnKF is attractive when building a new medium-range ensemble numerical weather prediction (NWP) system. However it is less suitable for NWP systems with uncertainty in a wide range of scales; it may not use high-resolution satellite data as effectively as 4D-Var. For limited-area mesoscale NWP systems a hybrid method is attractive.

KEYWORDS: Error covariance modelling Numerical weather prediction

1. INTRODUCTION

Numerical weather prediction (NWP) users are becoming more sophisticated in their ability to allow for the probable range of forecast errors when making decisions. Increased computing power has enabled the use of ensembles as a practical way of representing the probability distribution function (pdf) of errors, and of propagating that distribution in time. This paper discusses whether the ensemble representation of a pdf can be sufficiently accurate to enable near-optimal combination of forecast and observed information in an ensemble-based data assimilation scheme (e.g. Anderson 2001; Bishop *et al.* 2001; Houtekamer and Mitchell 2001; Evensen 2003), and compares the EnKF approach to the currently more popular variational approach. Variational data assimilation has been a very successful technique in operational NWP, first in a three-dimensional (3D) form (3D-Var, e.g. Parrish and Derber 1992; Courtier *et al.* 1998; Lorenc *et al.* 2000), then using an adjoint model to include the time dimension (4D-Var; e.g. Rabier *et al.* 2000). The key to its success is the emphasis on characterization of the error structures (and hence information content) of both the model forecast and the observations, enabling them to be combined in a nearly optimal way. This is important since model forecasts used in data assimilation are now typically as accurate as radiosonde observations (Simmons and Hollingsworth 2002), while satellite soundings have error characteristics which can be more easily represented in the direct variational assimilation of radiances (e.g. Andersson *et al.* 1994). The complete error structure of a time-evolving model trajectory requires a multi-dimensional pdf which is impossible to know or even represent accurately. Lorenc (2003) discusses how 4D-Var summarizes it by a covariance model including time-evolution within a short time-window by a linear perturbation model and its adjoint. The ensemble approach on the other hand represents the pdf less precisely, but can calculate its evolution over a longer time period.

Lorenc (2003) related 4D-Var to the Kalman Filter. In section 2 the basic ensemble Kalman filter (EnKF) method is reviewed in a similar notation. The EnKF is in principle simple to implement; this is discussed in section 4. But ease of implementation has to be weighed against effectiveness; in section 3 the effect of the EnKF covariances on the characteristics of the assimilation scheme is discussed. Although the standard EnKF, reviewed in section 2, is implemented as an observation-space algorithm, this is not

* Corresponding author: Met Office, FitzRoy Road, Exeter, Devon EX1 3PB, UK.

e-mail: andrew.lorenc@metoffice.com

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TABLE 1. SUMMARY OF THE COMPARATIVE EASE OF IMPLEMENTATION OF THE ENSEMBLE KALMAN FILTER (EnKF) AND 4D-VAR

	Incremental 4D-Var	EnKF
Forecast model	Predict evolution of mean. No switches.	Predict evolution of typical state. May have stochastic physics and switches.
Linear model	Predict average evolution of finite perturbations from the mean. May be simplified.	Not needed.
Adjoint model	Needed for (simplified) linear model.	Not needed.
Covariance model	Significant effort needed for covariance model, and for suitable linear approximations to nonlinear model. Adjoint code needed.	Little effort, other than to keep the ensemble spread matching the error.
Observation operators	Linear and adjoint operators needed, but these are not usually difficult.	Only uses forward operators.
Analysis algorithm	Descent algorithm available as 'off the shelf' software.	Little effort for a sequential algorithm such as the EnSRF. Simultaneous box algorithms are more complicated.
Suitability for parallel computers	Require parallel simplified and adjoint models.	Ensemble members can run on individual processors, but then sequential processing of observations requires inter-processor communication for each, and simultaneous processing in boxes requires a costly data transposition for the covariances.
Limited-area modelling	Error covariance models can be extended to specify boundary-value errors	Requires an ensemble of global forecasts to provide boundary conditions.

See section 4 for discussion.

essential. In section 5, I describe a hybrid scheme, using the ensemble covariance model in a standard model-space variational algorithm. This is less convenient for generating the ensemble, but might be a useful compromise if a forecast ensemble is generated by other methods. Finally, in section 6 I summarize sections 3 and 4 with the help of Tables 1 and 2, comparing the EnKF with variational methods and discussing the implications for developments to operational NWP assimilation systems.

2. REVIEW OF ENSEMBLE KALMAN FILTER METHOD

(a) Equations

The basic idea of the EnKF (Evensen 1994) is to construct an ensemble of forecast states $\{\mathbf{x}_i^f\}$, $(i = 1, \dots, N)^*$ such that the mean of the ensemble is the best estimate of the population mean, and the sample covariance is a good estimate of the forecast error covariance of this best estimate:

$$\mathbf{P}^f \approx \mathbf{P}_e^f = \overline{(\mathbf{x}^f - \overline{\mathbf{x}^f})(\mathbf{x}^f - \overline{\mathbf{x}^f})^T}, \tag{1}$$

* I follow the normal notation and use i as a subscript denoting the ensemble member. This should not be confused with its use to denote the time in the notation of Ide *et al.* (1997) and Lorenc (2003), for which in this paper I use k .

TABLE 2. SUMMARY OF THE ASSIMILATION CHARACTERISTICS OF THE ENSEMBLE KALMAN FILTER (EnKF) COMPARED WITH THOSE OF 4D-VAR¹

	Incremental 4D-Var	EnKF
Forecast covariances	Modelled at t_0 (usually isotropic), time evolution represented by linear and adjoint models.	Sampled by ensemble (flow-dependent). Noisy: must be modified to have compact support using Schur product.
Analysis method	Variational minimization using model increments, with full model outer loop.	Simple sequential method such as the EnSRF, or simultaneous observation space Kalman filter equation.
Ability to fit detailed observations.	Limited by resolution of simplified model.	Limited to fewer data (in a region) than ensemble members.
Balance constraints	Can be imposed through a dynamical design to the variable transform, or a separate balance penalty.	Only imposed if each forecast in the ensemble is balanced. Lost slightly in Schur product.
Nonlinear observation operators	Allowed if differentiable. (Results uncertain if pdf is bimodal in range of interest.)	Allowed, but resulting pdf modelled by Gaussian.
Non-Gaussian observational errors	Allowed if differentiable. (Results uncertain if pdf is bimodal in range of interest.)	Not allowed. Prior QC step is needed.

pdf is the probability density function.

¹Lorenc (2003).

where superscript T is the transpose, and the overbar denotes the ensemble estimate of the population mean. As discussed in Lorenc (2003), we follow the extended Kalman filter in representing the forecast error pdf by its mean and covariance, even though the forecast states are propagated in time by a nonlinear forecast model M . A similar approximation involving the nonlinear observation operator H allows us to calculate estimates of covariances used in the Kalman gain without needing its linearization $\mathbf{H}$:

$$\mathbf{P}^f \mathbf{H}^T \approx \overline{(\mathbf{x}^f - \overline{\mathbf{x}^f})(H(\mathbf{x}^f) - \overline{H(\mathbf{x}^f)})^T}, \quad (2)$$

$$\mathbf{H} \mathbf{P}^f \mathbf{H}^T \approx \overline{(H(\mathbf{x}^f) - \overline{H(\mathbf{x}^f)})(H(\mathbf{x}^f) - \overline{H(\mathbf{x}^f)})^T}. \quad (3)$$

These can be used in the standard Kalman analysis equation to update the best estimate (ensemble mean):

$$\overline{\mathbf{x}^a} = \overline{\mathbf{x}^f} + \mathbf{P}^f \mathbf{H}^T (\mathbf{H} \mathbf{P}^f \mathbf{H}^T + \mathbf{R})^{-1} (\mathbf{y}^o - \overline{H(\mathbf{x}^f)}). \quad (4)$$

The simplest way to generate the ensemble is to introduce stochastic perturbations into the forecast and analysis of each member:

$$\mathbf{x}_i^f(t_{k+1}) = M(\mathbf{x}_i^a(t_k)) + \boldsymbol{\eta}_i(t_k), \quad (5)$$

where $\boldsymbol{\eta}$ is a noise process with zero mean and covariance matrix $\mathbf{Q}$, and

$$\mathbf{x}_i^a = \mathbf{x}_i^f + \mathbf{P}^f \mathbf{H}^T (\mathbf{H} \mathbf{P}^f \mathbf{H}^T + \mathbf{R})^{-1} (\mathbf{y}^o - H(\mathbf{x}_i^f) + \boldsymbol{\varepsilon}_i), \quad (6)$$

where $\boldsymbol{\varepsilon}$ is a noise process with zero mean and covariance matrix $\mathbf{R}$. For a sufficiently large ensemble this EnKF gives the same error statistics as the standard Kalman filter (Burgers *et al.* 1998). Note that although (6) must be solved for each ensemble member, the term $\mathbf{P}^f \mathbf{H}^T (\mathbf{H} \mathbf{P}^f \mathbf{H}^T + \mathbf{R})^{-1}$ is common, and need only be evaluated once.

It is convenient to introduce a matrix notation for the ensemble, with $\mathbf{X}$ having as columns the normalized deviations from the ensemble mean:

$$\mathbf{X} = \frac{1}{\sqrt{N-1}}(\mathbf{x}_1 - \bar{\mathbf{x}}, \mathbf{x}_2 - \bar{\mathbf{x}}, \dots, \mathbf{x}_N - \bar{\mathbf{x}}). \quad (7)$$

These perturbation matrices are estimates of the square root of the covariances; for instance (1) can be replaced by

$$\mathbf{P}_e^f = \mathbf{X}^f \mathbf{X}^{fT}. \quad (8)$$

The EnKF is thus a square-root Kalman filter; the analysis method (6) allow us to calculate $\mathbf{X}^a$ from $\mathbf{X}^f$, and the forecast (5) allows us to propagate the square root of the covariance to the next time.

Instead of adding the noise in (6), it is possible to transform the ensemble $\{\mathbf{x}_i^f\}$ with covariance $\mathbf{P}^f$ into an ensemble $\{\mathbf{x}_i^a\}$ with the expected Kalman filter analysis error covariance $\mathbf{P}^a = \mathbf{P}^f - \mathbf{P}^f \mathbf{H}^T (\mathbf{H} \mathbf{P}^f \mathbf{H}^T + \mathbf{R})^{-1} \mathbf{H} \mathbf{P}^f$. First the ensemble mean is calculated using (4), and then the perturbation matrices are scaled to give the correct covariance. In the Ensemble Adjustment Kalman Filter (EAKF, Anderson 2001) an adjustment matrix $\mathbf{A}$ is implicitly calculated* by using a singular-value decomposition to find the square root of a matrix of order the number of observations, such that

$$\mathbf{X}^a = \mathbf{A}^T \mathbf{X}^f. \quad (9)$$

The Ensemble Transform Kalman Filter (ETKF, Bishop *et al.* 2001) requires the singular-value decomposition of a matrix of order N , to find matrix $\mathbf{T}$ such that

$$\mathbf{X}^a = \mathbf{X}^f \mathbf{T} \dots \quad (10)$$

Whitaker and Hamill (2002) demonstrated that the elimination of the sampling error associated with the perturbed observations improves the accuracy over methods with perturbed observations, for the same ensemble size. They presented an algorithm related to the EAKF called the ensemble square-root filter (EnSRF) which is particularly simple to apply when observations are processed sequentially, as discussed below. Tippett *et al.* (2003) show that these methods without perturbed observations are all forms of ensemble square-root filters, and compare and relate the ETKF, EAKF and EnSRF algorithms.

(b) Sequential application, or box method

Direct methods for analysing all observations together can be used, either by solving (6) iteratively as in PSAS (physical-space statistical analysis system, Cohn *et al.* 1998), or by converting it to a problem requiring singular-value decomposition of matrices of order N (Bishop *et al.* 2001; Tippett *et al.* 2003). However, most applications simplify the computations by splitting up the observations. The analysis step (6) (or the alternative (4) and (9)) can be applied sequentially to small batches of observations (Houtekamer and Mitchell 2001). The analysed ensemble $\{\mathbf{x}_i^a\}$ from one batch becomes the $\{\mathbf{x}_i^f\}$ for the next. As long as there is no error correlation between observations in different batches, this is equivalent to treating the observations in one large batch. This technique, together with the modification of the covariances to give compact support discussed in subsection 3(a), can be used to make the basic EnKF, or the EnSRF or EAKF, simple and computationally feasible for large systems.

* In practice the multiplication in (9) is performed without explicitly evaluating $\mathbf{A}$.

Alternatively, again helped by compactly supported covariances, the box method first introduced by Lorenc (1981) can be used. Instead of solving (6) for all the observations, the area is subdivided into boxes, and separate versions of (6) solved for the observations influencing each box (Keppenne and Rienecker 2002).

(c) Application to a time window

With appropriate changes to H to include an interpolation in time, all of the above equations still hold if we replace each 3D forecast state $\mathbf{x}$ by a 4D $\underline{\mathbf{x}}$ such that $\underline{\mathbf{x}}^T = (\mathbf{x}(t_1)^T \dots \mathbf{x}(t_k)^T)$. The covariances given by equations like (1) are then defined between all points in space and time spanned by $\underline{\mathbf{x}}$. This could allow the batching of observations into time windows, as in standard NWP assimilation cycles, without the error of assuming that all observations in a time window are simultaneous. (However, there is little advantage in this for the sequential schemes described in (b) above, and the costs of updating the ensembles increase with the size of $\underline{\mathbf{x}}$.) The time covariances are key to the use of the ETKF method for targeting adaptive observations (Bishop *et al.* 2001); they are used to calculate the decrease in error at the end of the time period due to various distributions of observations at the beginning.

(d) Inbreeding

For a small ensemble size, sampling errors in the covariance estimate can build up to give an ensemble whose covariance is too small. To counter this, Anderson (2001) adds an extra covariance inflation factor into (9), while Houtekamer and Mitchell (2001) split the ensemble into two, such that the analysis of one half is performed using the covariances estimated by the other. The covariances will also be too small if the model error term (with covariance $\mathbf{Q}$) is neglected in (5). Covariance inflation can compensate for this too; better methods are discussed by Mitchell *et al.* (2002); Evensen (2003) and Tippett *et al.* (2003).

3. ASSIMILATION CHARACTERISTICS OF ENKF

(a) The effect of noise in sampled covariances

If the ensemble is created by random sampling from a pdf, covariances estimated using (1) will have error proportional to $1/N$, as illustrated in Fig. 1, which shows sampled covariances from a Gaussian shaped covariance $\exp(-r^2/2a^2)$, with $a = 500$ km. A suboptimal Kalman gain calculated using the estimated covariances:

$$\mathbf{K}_e = \mathbf{P}_e^f \mathbf{H}^T (\mathbf{H} \mathbf{P}_e^f \mathbf{H}^T + \mathbf{R})^{-1}, \quad (11)$$

will give larger analysis errors, which can be calculated if one knows the true covariance:

$$\mathbf{P}^a = \mathbf{P}^f - 2\mathbf{K}_e \mathbf{H} \mathbf{P}^f + \mathbf{K}_e (\mathbf{H} \mathbf{P}^f \mathbf{H}^T + \mathbf{R}) \mathbf{K}_e^T. \quad (12)$$

Applying this to a global analysis using a single perfect observation and the covariances shown in Fig. 1 gives an average analysis error of $1 + 0.0087$, larger than the prior variance of 1.0. Perfect covariances (e.g. an infinite ensemble size in our example) reduce the global variance to $1 - 0.0015$. In other words, the harm done by spurious covariances with distant grid points is greater than the local benefit. One remedy is to carry out many approximate local analyses, only selecting local observations. This can introduce discontinuities in the analysis when an observation is selected for one grid point, but not for its neighbour. Houtekamer and Mitchell (2001) avoid discontinuities

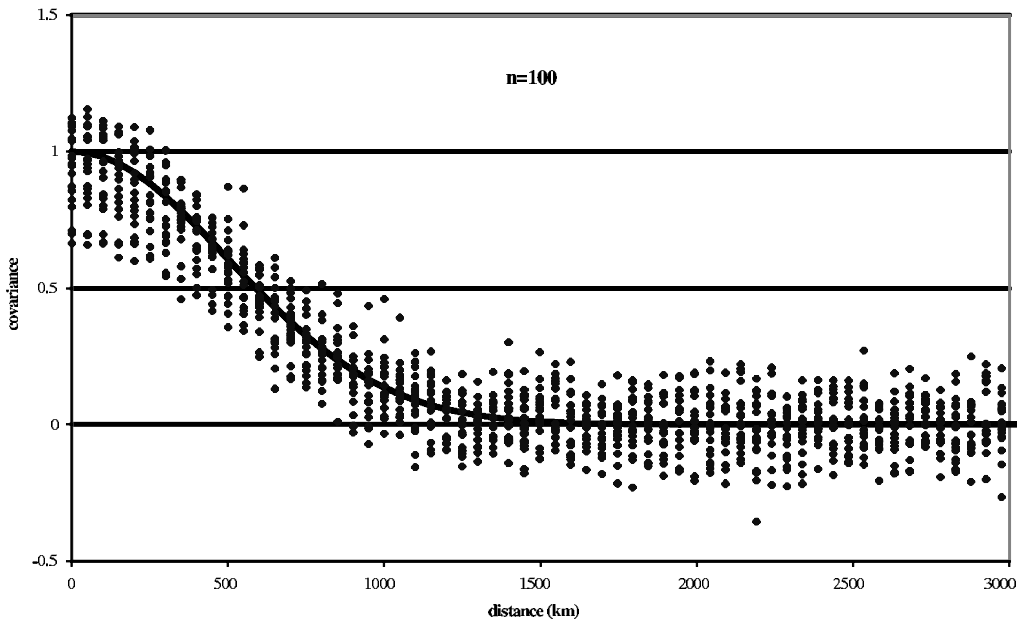


Figure 1. Ensemble estimate covariances with a point at the origin, for ensembles of 100 samples taken from a population whose covariance, shown in the solid curve, is $\exp(-r^2/2a^2)$, with $a = 500$ km (see text).

by modifying the covariances to make them ‘local’, using a Schur product*, with a correlation function with compact support. (The Schur product of two matrices having the same dimensions is denoted $\mathbf{A} = \mathbf{B} \circ \mathbf{C}$ and consists of the element-wise product such that $A_{i,j} = B_{i,j}C_{i,j}$. If $\mathbf{B}$ and $\mathbf{C}$ are valid covariance matrices, then so is $\mathbf{A}$.) I follow them, and use as $\mathbf{C}$ the correlation function in Eq. (4.10) from Gaspari and Cohn (1999). This requires a scale parameter c ; the correlations go to zero at distance $2c$. This is applied as a simple function of horizontal distance, to all levels and variables. Gaspari and Cohn (1999, their Fig. 6) show that their Eq. (4.10) is very similar to the Gaussian shaped covariance assumed in Fig. 1. By trying different values of c , and calculating the analysis error for each using (12), an optimal value can be found. The results are shown in Fig. 2, for different ensemble sizes. The scales are shown relative to that which matches the curve in Fig. 1: $c = a(0.3)^{1/2} = 913$ km, since then results are not sensitive to the value chosen for a . The covariances modified with the Schur product using the optimal c are shown in Fig. 3. Hamill *et al.* (2001) reach a similar conclusion that the use of covariance localization is beneficial when the ensemble size is small.

The use in $\mathbf{C}$ of a compact support function which is zero beyond a known distance allows significant computational savings, for instance in the $\mathbf{P}^f \mathbf{H}^T$ term of (6) and the $\mathbf{A}^T$ term of (9). It also helps justify the box method discussed in subsection 2(b). The ETKF method transforms the matrix problem to that of determining $\mathbf{T}$ in (10) of order the ensemble size N , rather than working in the space where $\mathbf{C}$ is defined. Because of this, it is not possible to include a Schur product in the ETKF method as currently defined.

* Also known as a Hadamard product.

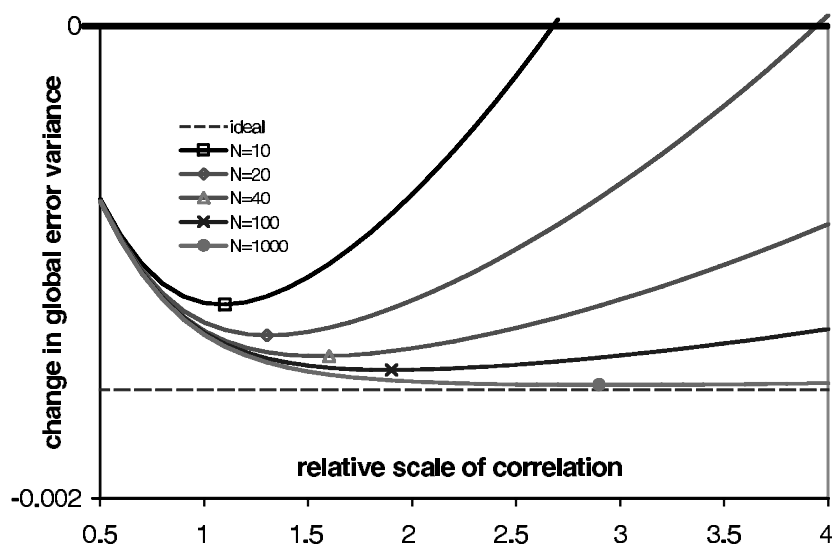


Figure 2. The change in the global average error variance, calculated using (2), for ensemble covariances like those in Fig. 1, modified by a Schur product with a correlation C . The change is plotted as a function of the relative scale of the correlation function C , such that 1.0 implies that C is similar to the covariance curve shown in Fig. 1. A symbol marks the best scale for C (by this measure), different curves are for different ensemble sizes, N , and the dashed line shows the error reduction achievable with perfect covariances.

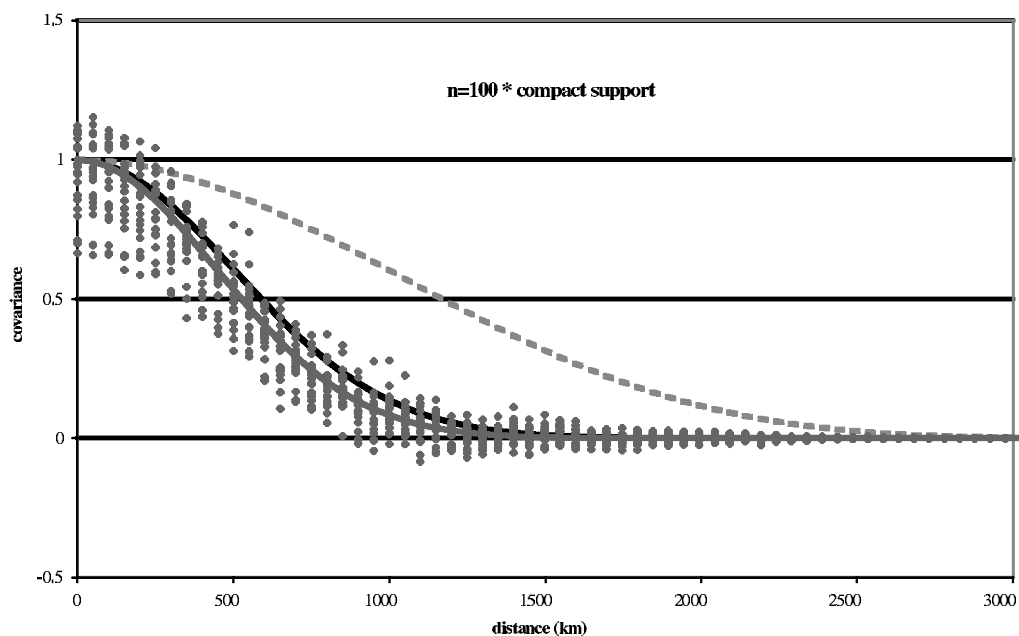


Figure 3. As Fig. 1, with the ensemble covariances modified with the Schur product correlation C with scale corresponding to the symbol in Fig. 2 (shown dashed). The Schur product of C with the population covariance is shown by the lower grey line.

(b) *The effect of degenerate covariances*

The basic EnKF (without the Schur product) only has N degrees of freedom available to fit the observations. This is immediately obvious from the variational formulation of section 5, and from (4) or (10) which have matrices of order N . It is perhaps less apparent from the sequential algorithm (although this can be shown to be equivalent to a simultaneous analysis); appendix A shows that a perfect observation removes a degree of freedom from the ensemble. Thus the covariance given by (1) has at most N non-zero eigenvalues, reduced by one for each recent accurate observation. Although, in theory, positive definite observational errors $\mathbf{R}$ keep equations well-behaved, in practice unrealistic extrapolations into data-sparse areas can result. The EAKF and ETKF algorithms perform singular-value decompositions, and are well-behaved if vectors with small singular values are omitted (Anderson 2001). It is clear from section 5 below, that the Schur product adds extra degrees of freedom. Houketer and Mitchell (2001) say that the Schur product greatly improves the conditioning, implying that it was sufficient to keep the system well-behaved for their tests.

A consequence of this degeneracy is that the ensemble Kalman filter is unable to fit more than N data within any area where the effect of the Schur product is insignificant (i.e. where $\mathbf{C}$ is close to 1). Given the limitations on the scale of the Schur product implied by Fig. 2, this is a serious limitation for some applications. Of particular concern is the ability to fit observations determining the moisture field. Moisture fields often have more horizontal detail than the dynamical fields upon which EnKF experimentation has so far concentrated, and there are observations available (from satellites and radar) which give frequent detailed information about them. As correlations between temperature and moisture are important they should be analysed together. The horizontal scale of the Schur product must be larger than the scale of the dynamical errors that can affect temperatures even in a mesoscale model, making it difficult to fit many details observed in the moisture field on a smaller scale. (This problem might be alleviated by having a more complicated form for $\mathbf{C}$ than the simple function of horizontal distance used so far. But it may be difficult to specify such a $\mathbf{C}$; the theoretical basis of the EnKF is compromised when $\mathbf{C}$ has a significant affect of the results, rather than mainly reducing noise and computer costs.)

(c) *Balance*

The atmospheric state is attracted towards a slow manifold, with flow close to balance which can be approximated by the geostrophic relationship. As long as the noise introduced in each analysis step is less than the dissipation of imbalance during the forecast, the states $\{\mathbf{x}_i^f\}$ will be close to balance, and this will be reflected by the covariances in (1). In assimilation schemes with local data selection (e.g. Lorenc 1981), changes in data selection from point to point introduce more imbalance than is usually dissipated during a 6-hour forecast to the next batch of observations. It has, therefore, been usual to include an initialization step in the data assimilation cycle. The Schur product reduces this data selection noise (Houketer and Mitchell 2001), but it still introduces imbalance. Let us assume that the ensemble $\{\mathbf{x}_i^f\}$ is in geostrophic balance, with geopotential height error covariance as in Fig. 1. Then, without the Schur product the height increments from a single height observation will approximate the curve in Fig. 1, and the wind increments will be in geostrophic balance with its gradient. With the Schur product the height increments will drop off more rapidly with distance, as shown in Fig. 3, requiring an increased wind increment to balance them, while the actual wind increments implied by the ensemble covariance are reduced by the Schur product factor.

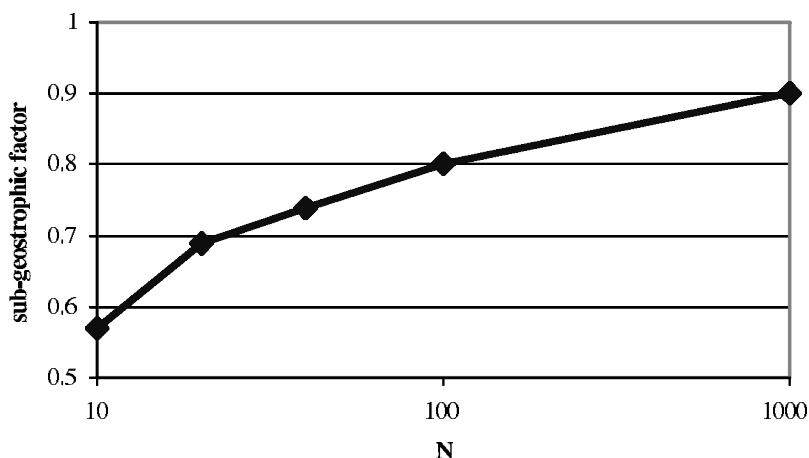


Figure 4. Factor by which, because of the Schur product, the analysed wind will be less than that required to balance the analysed height gradient, at 500 km from a single height observation, as a function of ensemble size, N . The best scales for correlation C , as shown in Fig. 2, were used.

Figure 4 quantifies this effect; the wind increments implied by a height observation are significantly sub-geostrophic. This result suggests that the EnKF may need either an explicit initialization step or the use of a forecast model with extra damping of gravity waves. This would make it harder to get reliable short-period forecasts of precipitation. Mitchell *et al.* (2002) investigated the effect of covariance localization in an EnKF using a dry, global, primitive-equation model. They found that severe localization did cause substantial imbalance, but for their system the imbalance due to a localization radius greater than about 3000 km was not significant.

In variational assimilation methods, balance is usually introduced through additional ‘slow’ penalties (Gauthier and Thépaut 2001), or explicitly through the design of the covariance model. Lorenc (2003) discussed how the balance constraints are imposed via a null space of the covariance model. As discussed in the previous section, an EnKF covariance has a spurious null space due to the limited ensemble size. It is impossible to relax this by the Schur product, or otherwise, without relaxing the balance constraints at the same time. Additional initialization is therefore more likely to be necessary.

(d) *Non-Gaussian errors and nonlinearities*

The equations derived in subsection 2(a) are only optimal if the error pdfs are Gaussian*. Nonlinearities in M and H , and the possibility of gross observational errors, can all give non-Gaussian pdfs. Since we have an ensemble to represent the forecast pdf, it is possible to derive alternative methods which do not assume it to be Gaussian. Similarly, since we know the properties of H and of the observational errors, it is possible to allow for a non-Gaussian observational error pdf. However, although possible in principle, in practice there are serious difficulties. Some assumption must be made about the forecast error pdf sampled by the ensemble; none is obviously best. For the sequential processing to be optimal, the ensemble has to represent accurately

* The equations can also be derived as the best linear unbiased estimator without making assumptions about the form of the pdfs, but for non-Gaussian pdfs the optimal equations should be nonlinear.

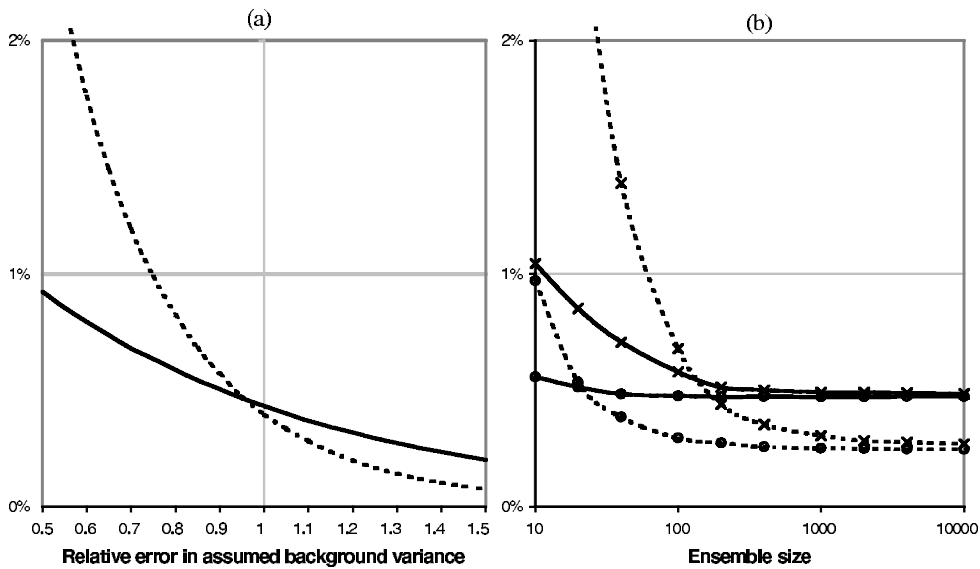


Figure 5. Percentage of all observations correctly identified as having gross errors (solid lines), and falsely identified (dotted lines): (a) using the Bayesian quality-control algorithm of Lorenc and Hammon (1998), for a range of assumed background error variances; (b) lines with o use the variance from an ensemble in the same algorithm, and lines with x use the ensemble probability distribution function directly with no assumption about its shape, for a range of ensemble sizes.

the key features of the intermediate pdf. For instance, if the observation being processed is close to being rejected as erroneous, then the pdf will become bi-modal, and the ensemble should have members distributed in each peak according to its probability. The problems of sampling error seen in subsection 3(a) will be magnified. (This is illustrated by the lines with x symbols in Fig. 5, discussed in the next section). Then if a subsequent observation is processed which supports the first, one mode becomes much more likely. If we are to retain the assumption that the ensemble members are equally likely, then some must be moved from one peak to another. This is not easy to achieve; the linear mapping of (9) or (10) has to be replaced by a nonlinear re-sampling. Such methods have been demonstrated successfully for simple low order models (Pham 2001; Anderson 2003). But for the large systems and relatively small ensembles used in NWP it is not currently practicable to replace the main methods of subsection 2(a). Some allowances might still be made for non-Gaussian errors in pre-processing, as discussed in the next two subsections.

(e) Quality control

The properties of observations which may have gross errors can be represented by non-Gaussian distributions with long tails (Lorenc and Hammon 1988, hereafter L&H). Variational methods can cope with these directly (Dharssi *et al.* 1992, Andersson and Jarvinen 1999). These methods allow an implicit choice to be made as to whether to believe an outlying observed value, dependent on whether it is consistent with surrounding values. In the algorithms of subsection 2(a) this is not possible, since the observational error pdfs are only summarized by their covariances.

It is still possible to use the ensemble information in a prior quality-control step. Many critical quality-control decisions are made in data-sparse areas, where the principal source of corroborative information is the forecast. If the forecast error variance modelled by $\{x_i^f\}$, despite sampling noise, is more accurate than the error variances assumed in variational schemes, then the ensemble covariances can do better. Figure 5 illustrates this using values typical of ship pressure observations taken from L&H. A million observations were generated, 97% of them with Gaussian errors with variance 1 hPa^2 , and 3% with ‘gross errors’ in which case they can have any plausible value with a constant pdf of 0.043 hPa^{-1} . Note that we are assuming prior removal of extreme observations; the bad observations only have errors in the range $\pm 11.6 \text{ hPa}$. So a significant number of the gross errors are in the same range as good observations, and cannot easily be detected. L&H assumed forecast errors were Gaussian, with variance 3.6 hPa^2 . It is more realistic to allow for day-to-day changes in the variance, so forecast values were calculated for each observation, sampled from a range of Gaussians with variances spread evenly from 1.8 to 5.4 hPa^2 . Figure 5(a) shows the result of applying the quality-control check of L&H Eqs. (20) and (21), using the same assumed forecast error variance for all the observations. Figure 5(b) shows the results if the forecast error distribution is predicted by a perfect ensemble (i.e. the ensemble members were created by random sampling from a Gaussian with the correct variance for that observation’s position). The lines with o symbols were obtained by putting the ensemble variance into the same L&H quality-control equations. Even with an ensemble as small as 40 members, the use of an ensemble estimate of the variance for each observation leads to better decisions than just using the average variance*. As discussed in the previous section, an ensemble can represent a non-Gaussian forecast pdf. A Bayesian quality-control algorithm can use the ensemble directly, replacing the integral over the pdf by a sum over ensemble members. The observation is rejected if

$$\sum_{i=1}^N P(y^o | x_i \cap \overline{G}) P(\overline{G}) < \sum_{i=1}^N P(y^o | x_i \cap G) P(G). \quad (13)$$

The left-hand side is evaluated assuming the observation does not have a gross error ($\overline{G}$), using the Gaussian observational error distribution, and the right-hand side just sums the constant pdf (0.043 hPa^{-1}) multiplied by the prior probability of a gross error (0.03). The results of this method are shown in Fig. 5(b) by the lines with x symbols. As expected, many more members are needed to adequately define the pdf if no assumptions are made about its shape; in this case at least 400 members are needed for good results. (This result supports the conclusion of subsection 3(d).)

Checks such as this could easily be incorporated into a sequential EnKF, deciding whether to accept or reject each observation just before using it. But in this example only about one sixth of the erroneous observations could be detected—even a good knowledge of its variance cannot compensate for the forecast not being very accurate. Using nearby observations in a ‘buddy-check’ is more reliable (L&H). Unfortunately such methods require all the observations to be processed together, while in a sequential EnKF algorithm early observations influence the quality control of later observations, but not vice versa. So if the ensemble covariances are used in a quality-control step

* This work only considers the benefits of ensemble predictions of covariances. It is possible that an ensemble would also provide a more accurate prediction of the mean, but this has not been demonstrated for short forecasts. Hence I am assuming the forecast value used in the conventional quality control of Fig. 5(a) equals the population mean in the ensemble methods of Fig. 5(b).

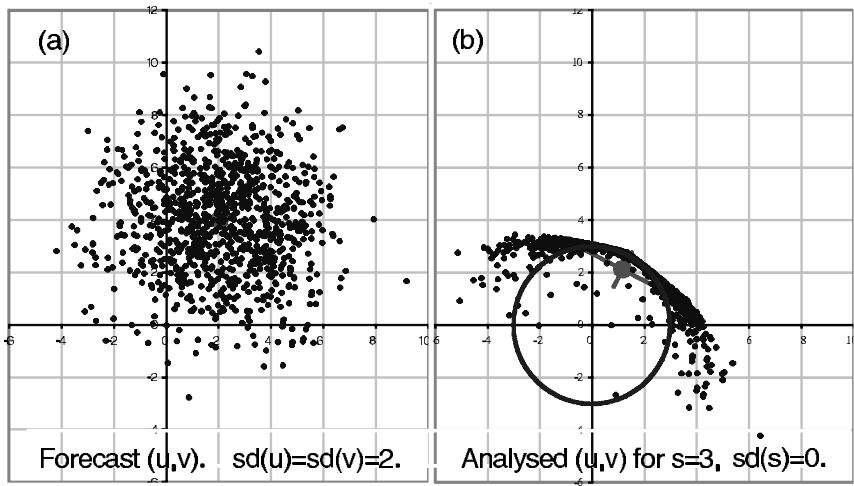


Figure 6. (a) A 1000 member ensemble of forecast wind vectors, with mean (2, 4) and standard deviation (2, 2); (b) an ensemble of wind vectors resulting from an ensemble Kalman Filter analysis of an error-free observation of wind speed; the circle shows the observed value, which is 3. In each frame the large grey dot shows the mean of that distribution; whilst in (b) the grey lines show its eigenvectors.

within a sequential EnKF analysis algorithm, results will be sub-optimal*. On the other hand it is possible to incorporate quality control into the simultaneous processing of observations partitioned into boxes as described in subsection 2(b) (Lorenc 1981).

(f) Nonlinear observation operators

Nonlinear relationships between the observed quantities and the model variables are allowed for in the EnKF by (2) and (3); in effect an average linearization across the whole prior pdf is used. Quantities such as rainfall which are diagnosed during the model forecast are allowed for by augmenting the $\{\mathbf{x}_i^f\}$ to include a rainfall rate variable. This makes it easy to design an EnKF which uses such observations without understanding and linearizing the processes involved, as would be necessary for a variational method.

However, even if the forecast error pdf is Gaussian, and can be described by $\mathbf{P}^f$, the action of nonlinear $\mathbf{H}$ gives a non-Gaussian distribution. The use of (2) and (3) does not allow for this, while variational methods with a non-quadratic penalty function can. The simplest example is an observation of a wind speed. Figure 6(a) shows an approximately Gaussian forecast ensemble of wind vectors, and Fig. 6(b) the resulting analysis ensemble. The joint pdf of $\mathbf{x}$, with components (u, v) , and wind speed $s = H(\mathbf{x}) = (u^2 + v^2)^{1/2}$ is non-Gaussian, and therefore not fully described by the covariance obtained by substituting this $\mathbf{x}$ and $\mathbf{H}$ in (2), making the analysis using (6) sub-optimal. Hence the analysed ensemble does not represent the ideal posterior pdf, which should lie on the circle. In the subsequent analysis of other observations, this ensemble will be represented by the Gaussian whose mean and eigenvectors are the grey dot and cross. This sub-optimality probably does not matter for a single wind speed observation, particularly when realistic observation errors are included; but it

* If the ensemble is large enough to properly represent the tails of a non-Gaussian pdf, and if the nonlinear resampling methods mentioned in subsection 3(d) are used, it is theoretically possible to construct an optimal sequential algorithm.

might matter for some dense highly nonlinear observations, such as scatterometer winds, where the detailed information in nearby observations can combine synergistically. Variational methods, allowing for nonlinear observation operators, accurately represent the pdf in the vicinity of the latest best estimate; in our example they would always give an analysed wind on the circle. They have proved useful (Stoffelen and Anderson 1997; Isaksen and Stoffelen 2000), allowing the analysis to select implicitly between wind aliases to give a horizontally consistent pattern.

4. EASE OF IMPLEMENTATION

(a) *Choices within an EnKF*

For the sizes of ensemble currently envisaged for NWP, subsection 3(a) showed that covariance localization is necessary. The ETKF algorithm cannot easily do this, so we do not consider it further for application to NWP data assimilation.

The next choice is whether to perturb the observations. The original implementation by Evensen (1994) did not perturb the observations, but this was shown by Burgers *et al.* (1998) to be incorrect, and the ‘basic’ EnKF of (6) includes perturbed observations. Whitaker and Hamill (2002) showed, however, that methods which adjust the ensemble instead (EAKF, ETKF or EnSRF) are more accurate for a given ensemble size. Excluding the ETKF, all such methods currently published are only practicable with sequential processing of observations.

The sequential processing of observations described in subsection 2(b) can be used in the EnKF with perturbed observations (Houtekamer and Mitchell 2001), or in the EnSRF (Whitaker and Hamill 2002). The approach has an attractive simplicity; however, as discussed in subsection 3(e), it is less able to cope with quality-control decisions. Alternatively, the simultaneous processing of observations in (6) can be well approximated using a box method, as shown for an ocean model by (Keppenne and Rienecker 2002), and being developed for global NWP in Australia (P Steinle, personal communication).

(b) *Forecast model*

It is possible to use a forecast model in the EnKF which represents all processes thought to be important. Each ensemble state represents a possible atmospheric state, so conceptually simple physical ideas can be used in the design of each parametrization, and of the stochastic forcing terms representing uncertainties in the model formulation. No linearization, or adjoint model, is needed.

In comparison the full model used in 4D-Var should evolve the ensemble mean, which may not be a plausible physical state, and parametrizations should allow for the effect of perturbations about this state (Lorenc 2003). Fast processes (compared to the time window of the 4D-Var) whose timing is uncertain, are particularly hard to treat. In practice, smoothly evolving forecast models do not exist, and approximations have to be made in the linearization and adjoint. Processes which have proved difficult include boundary-layer fluxes, and evaporation of falling precipitation—both of which can proceed rapidly and then saturate. There is no problem in including a straightforward physical parametrization of these processes in the EnKF.

(c) *Covariances*

A key part of a variational data assimilation system is the background error covariance model; considerable effort is expended designing and coding this. The EnKF

is simpler; the major design problem is to keep the ensemble covariance matching that of the error.

(d) *Observation operators*

In comparison to that for the forecast model, little effort is needed to produce linear and adjoint versions of the observation operators for variational schemes. The effort is probably worth it for the extra flexibility variational schemes have in dealing with nonlinear observations (discussed in subsection 3(f)). The EnKF only needs the full observation operators, linearizations are calculated implicitly in (2) and (3). These implicit linearizations are fitted over the entire prior ensemble; the EnKF is inferior in this regard to 4D-Var which linearizes explicitly about the current best estimate. An exception to this might be precipitation; it would be attractive to be able to assimilate precipitation observations in an EnKF without needing to worry about linear adjoints. (But an EnKF using precipitation observations has yet to be demonstrated.)

(e) *Analysis algorithm*

Sequential algorithms such as the EnSRF, localized using a Schur product, are very easy to build. However, they are not as easily adapted as 4D-Var to incorporate quality control of the observations. Moreover, the sequential algorithm may not be efficient on a parallel computer if there are many observations, as discussed below. The alternative is a simultaneous solution of the observation-space analysis equation (6)—note that even this becomes expensive for large numbers of observations. It can be tackled using a box method in a straightforward way (Keppenne and Rienecker 2002). Or one could find global solutions to (6) using an iterative PSAS-like algorithm (Cohn *et al.* 1998), but the complexity of a computationally efficient algorithm would obviate one advantage of the EnKF—its simplicity.

Once the other components (e.g. the adjoint model) are available, the 4D-Var analysis algorithm only needs a descent algorithm, which is available as an ‘off the shelf’ package.

(f) *Parallel computing*

If 4D-Var uses $O(100)$ iterations running its linear and adjoint model, and the EnKF uses $O(100)$ ensemble members, their computational costs should be similar. The 4D-Var runs have to be performed sequentially, so the linear and adjoint models have to be designed to parallelize on multi-processor computers. In practice this limits the number of processors that can be used effectively. For the EnKF there is more choice. An obvious approach is to do the forecasts in parallel, each on their own processor or group of processors. This leaves the states distributed inconveniently for the calculation of covariances such as (2). An algorithm such as the EnSRF which processes observations sequentially is inherently difficult to parallelize—it would need significant inter-processor communication for each observation, making this approach impracticable if there are many observations. If the observations are processed simultaneously, the calculation of the covariances can be facilitated by a transposition of the ensemble matrix, which can be expensive (Keppenne 2000). To avoid this, Keppenne and Rienecker (2002) adopt a similar parallelization strategy to 4D-Var, with each model run distributed over all the processors, so that the box method solution of (6) can proceed in parallel with the minimum of extra inter-processor communication. However, because of the memory required on each processor, this approach is not easily scalable to large ensembles.

(g) *Limited-area modelling*

In order to provide high resolution over an area of interest, most NWP centres run limited-area forecast models, with lateral boundary conditions specified from a global model. Covariance models in variational schemes can be extended to specify plausible errors in these. Despite theoretical shortcomings in the specification of boundary conditions, in practice such NWP systems give useful short-range forecasts, away from the boundary. This is because the covariances used in the important central area are directly specified, and care can be taken that they are appropriate. EnKF methods however depend on the ensemble members evolving realistic error structures over an extended assimilation cycle; the covariances are indirectly specified by the sources and growth of perturbations in the ensemble. For a small area, during the time taken for small perturbations to grow to significant size, information can propagate across the boundary to affect the entire area. Forecast errors will be more influenced by boundary perturbations than by analysis perturbations and error growth within the area. So proper specification of an ensemble of boundary values, characterizing the incoming perturbations, would be essential for a limited-area EnKF. Probably this requires an ensemble of global models providing the boundary conditions.

Although research is underway into the use of limited-area ensembles for forecasting, it has concentrated on perturbing the boundary values and the model formulations, with no published work on the EnKF.

5. VARIATIONAL METHODS USING ENSEMBLE COVARIANCES

(a) *Variational fit to ensemble perturbations*

It is possible to use the ensemble covariance (8), and its modified version using a Schur product, in the background term of a variational analysis scheme (either 3D-Var or 4D-Var). Note that the 4D-Var method, as presented in Lorenc (2003), is designed to estimate the ensemble mean. The method proposed here is simply modifying that approach to use ensemble covariances; it does not provide a means of generating an analysed ensemble for a cycling EnKF. The ensemble is envisaged to come from another system which generates perturbations, centred on the ensemble mean from 4D-Var. As this ensemble system is mainly intended to improve the covariances used in 4D-Var, rather than directly assimilate all the information in observations, a cheaper solution such as a low-resolution ETKF should be sufficient.

As discussed in subsection 3(b), $\mathbf{P}_e^f$ is degenerate, so we cannot immediately use it in a standard variational penalty function for $\mathbf{x}$. The null space of $\mathbf{P}_e^f$ constrains $\mathbf{x}$ to stay in the space spanned by the ensemble. Lorenc (2003) discussed how to incorporate such constraints by transforming a simpler control variable. In this case I choose the transformation from a vector $\boldsymbol{\alpha}$ of N coefficients α_j :

$$\mathbf{x} = \overline{\mathbf{x}}^f + \mathbf{X}^f \boldsymbol{\alpha}. \quad (14)$$

Then, if the covariance $\mathbf{B}_{(\boldsymbol{\alpha})} = \langle \boldsymbol{\alpha} \boldsymbol{\alpha}^T \rangle = \mathbf{I}$, the covariance of the $\mathbf{x}$ defined by (14) satisfies (8). So the variational analysis for the transformed variable $\boldsymbol{\alpha}$ is obtained by minimizing the squared control variables plus the misfit between the predicted values $\mathbf{y}$ and observations $\mathbf{y}^o$:

$$\begin{aligned} J(\boldsymbol{\alpha}) &= \frac{1}{2} \boldsymbol{\alpha}^T \boldsymbol{\alpha} + \frac{1}{2} (\mathbf{y} - \mathbf{y}^o)^T \mathbf{R}^{-1} (\mathbf{y} - \mathbf{y}^o) \\ \mathbf{y} &= H(\overline{\mathbf{x}}^f + \mathbf{X}^f \boldsymbol{\alpha}). \end{aligned} \quad (15)$$

(b) *Using the Schur product to reduce degeneracy*

In order to match the covariance modified by the Schur product, we use as control variable a matrix α whose columns are N fields α_i , each with the desired modifying covariance $\mathbf{C}$. The variational problem is then to minimize

$$J(\alpha) = \frac{1}{2} \alpha^T \begin{pmatrix} \mathbf{C} & & 0 \\ & \ddots & \\ 0 & & \mathbf{C} \end{pmatrix}^{-1} \alpha + \frac{1}{2} (\mathbf{y} - \mathbf{y}^o)^T \mathbf{R}^{-1} (\mathbf{y} - \mathbf{y}^o)$$
$$\mathbf{y} = H \left\{ \overline{\mathbf{x}}^f + \mathbf{X}^f \circ \alpha \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \right\}, \tag{16}$$

where we use a similar Schur product* in calculating $\mathbf{x}$ and then $\mathbf{y}$ (see appendix B). As in standard 3D-Var methods (e.g. Lorenc *et al.* 2000), the inversion of the block diagonal matrix is avoided by a further horizontal transform of each field α_i into spectral space. Note that where (14) multiplies each ensemble member by a scalar coefficient, (16) has a Schur product with a horizontal field of coefficients. This increases the degrees of freedom available to the variational algorithm. The fields α_i are constrained to be smooth, with the horizontal scale in $\mathbf{C}$ from two to three times that of forecast error covariances (Fig. 3). In response to a single observation, the fields α_i will be proportional to the correlation function, dropping to zero away from the observation.

(c) *Hybrid variational plus ensemble covariances*

We could, in principle, minimize an equation like (16) N times, with perturbed observations, in the equivalent of (6), but this would be less efficient than the traditional EnKF method in generating an ensemble, since the saving obtained through the common term in (6) is not readily available. The main advantage is that we can have a hybrid approach, with the traditional variation control variable v (Lorenc 2003) supplemented by α . The incremental variational method can be used, where the control variables and the ensemble are at reduced resolution and $\mathbf{y}$ is given by equation (35) of Lorenc (2003). In a variational algorithm, whatever sequence of operations we use for calculating $\mathbf{y}$, we need its linearization and adjoint to carry out the minimization of

$$J(\mathbf{v}, \alpha) = \frac{1}{2} \mathbf{v}^T \mathbf{v} + \frac{1}{2} \alpha^T \begin{pmatrix} \mathbf{C} & & 0 \\ & \ddots & \\ 0 & & \mathbf{C} \end{pmatrix}^{-1} \alpha + \frac{1}{2} (\mathbf{y} - \mathbf{y}^o)^T \mathbf{R}^{-1} (\mathbf{y} - \mathbf{y}^o)$$
$$\mathbf{y} = H \left\{ \mathbf{x}^g + (1 - \beta) \mathbf{U} \mathbf{v} + \beta \mathbf{X}^f \circ \alpha \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \right\}, \tag{17}$$

where, as noted above, a further horizontal transform of α removed the need to invert $\mathbf{C}$. The terms $(1 - \beta)$ and β are included because the two control variables are really alternatives; when adding the α control variable one should reduce the use made of the

* In most implementations to date $\mathbf{C}$ is constructed using a single horizontal correlation function, defining the correlation between any variables at any levels to be a function of position only. The vertical and inter-variable covariances are left as specified by the ensemble—in effect $\mathbf{C}$ is constructed with these correlations all set to one. In a similar way the α fields used in the Schur product in (16) are built up by replicating a single 2D field of coefficients. In the control variable and first penalty term we need only consider the single horizontal field.

traditional control variable*. The ensemble perturbations can come from another source such as an error breeding cycle (Toth and Kalnay 1997). It has proved relatively straightforward to code this within the Met Office variational assimilation, with encouraging initial results (Dale Barker and Adrian Semple, personal communication). A similar hybrid method, not using the α control variable and localizing Schur product, has been demonstrated on a simple model by Hamill and Snyder (2000).

The hybrid method is easily extended from 3D-Var to 4D-Var by inserting a forecast step into the calculation of y , as in normal 4D-Var schemes. Note that in this case the ensemble members are still used at a single time at the beginning of the window; the 4D covariances discussed in subsection 2(c) are not used. Within the window the time evolution of covariances is modelled by 4D-Var, and the EnKF models their evolution from one window to the next.

The variational methods in this section do not make the approximations (2) and (3), and so are able to use nonlinear observation operators, and variational quality-control methods which simultaneously check all observations, while using covariances (partly) modelled by the ensemble, giving the advantages discussed in subsection 3(e).

6. DISCUSSION AND CONCLUSIONS

In performing this review it has become apparent that there is not a single EnKF algorithm to compare to 4D-Var. Even if one supports the conclusion of subsection 3(a), and rules out methods without covariance localization†, two extremes can be identified: sequential algorithms such as the EnSRF (Whitaker and Hamill 2002), and simultaneous algorithms such as the box method of Keppenne and Rienecker (2002). The method of Houtekamer and Mitchell (2001) is intermediate. In section 4, I compared the easy method of implementation of the EnKF and 4D-Var (summarized in Table 1), concluding that, for some aspects, the design of an EnKF scheme is simpler. For ensemble sizes $O(100)$ costs should be similar, as long as the number of observations is sufficiently small so that the costs of their processing and analysis are small compared to the ensemble forecasts. So using an EnKF is attractive for a centre which does not have existing 4D-Var infrastructure. It is also attractive in generating an ensemble which will be useful in providing probability forecasts to users.

But this attractiveness will be lost if better estimates of the forecast errors are at the expense of degradation to the best estimate analysis and forecast, so the characteristics of the EnKF discussed in section 3 are crucial. Table 2 summarizes these and compares them with those of 4D-Var.

Figure 2 shows that the deleterious effect of sampling noise on the covariances can be ameliorated by an appropriate Schur product for an ensemble size of $100 \sim 1000$. Current forecast models represent well the dynamical growth of midlatitude weather systems over periods of a few days, and the EnKF should model the consequential error covariances, which depend on the synoptic flow, better than 4D-Var (which is limited to modelling the linear evolution within the 4D-Var window). So for the midlatitude medium-range forecasting problem, where such weather systems are important and where ensemble predictions have proven utility, the EnKF is attractive. Designs with sequential processing of observations are simpler, but they are not able to include quality

* In practice one might want to make β scale-dependent, so that small scales are determined by the traditional control variable, while synoptic scales rely more on the ensemble.

† Evensen (2003) and Bishop (personal communication) argue that the imbalance introduced by localization, and the advantage of efficient simultaneous processing of observations, outweigh the problems discussed in subsection 3(a).

control—a pre-processing step is needed. Practical forms of EnKF cannot use optimally nonlinear observations with non-Gaussian pdfs. But the methods can use such data simply and directly in a sub-optimal way; alternatively, a separate pre-processing step might use the ensemble covariances, for instance for quality control and scatterometer de-aliasing (subsections 3(d), (e) and (f)). The ensemble should usefully predict day-to-day variations in the (co)variances used in pre-processing these observations, but it is unlikely the ensemble will be large enough to accurately represent non-Gaussian properties of the forecast errors (Fig. 5(b)). A remaining worry for the application of the EnKF to medium-range forecasting is that, even for synoptic-scale forecasts, important information may be present in observations which require high-resolution assimilation, for instance the tracer observations discussed below. Improved assimilation of satellite radiance in a high-resolution 4D-Var system has been an important factor in recent improvements in medium-range forecasts (Simmons and Hollingsworth 2002); the use of the same data in an EnKF has yet to be demonstrated.

There are practical limitations on the number of observations imposed by the parallel-computing costs of the sequential algorithm, or by the matrix sizes in the observation-space algorithm. More fundamentally, a finite ensemble size limits the ability to fit detailed observations. These concerns may be addressed by combining dense observations into fewer ‘super-obs’ (Lorenc 1981; Keppenne and Rienecker 2002). But the inability to fit detailed observations may be a significant problem for some applications. The atmosphere has many superimposed scales of motion; even for mesoscale assimilation systems synoptic-scale errors occur, and the dynamical error covariance scales are several hundreds of kilometres. The scale of the Schur product correlation should be two to three times this to get accurate dynamical analyses, as shown in Fig. 3. But the scales which we wish to analyse in humidity fields are much smaller, and from satellite and radar we may have many hundreds of observations within the Schur correlation radius. Providing a good fit to these observations is important for at least two reasons:

- to have accurate high-resolution cloud and precipitation at the beginning of a short-period mesoscale forecast (if the ‘nowcast’ does not correctly represent the current weather, customers lose confidence in the forecast);
- to use the information from the movement seen in consecutive observations to deduce advecting velocities.

This latter effect has been demonstrated in 4D-Var mesoscale assimilation of radar reflectivity, and in global 4D-Var assimilation of satellite soundings. The Kalman filter is similarly capable of extracting information about the advective velocity (Daley 1996), but only if the analysis fields can represent the pattern seen in the first set of observations. To extract this tracer information, the assimilation needs to process frequent observations. The EnKF may also have some practical disadvantages in this, because of the need for initialization (discussed in subsection 3(c)), and also the transposition costs (discussed in subsection 4(f)).

Most NWP systems of the future will have to assimilate remotely sensed observations with high spatial and temporal resolution to produce detailed analyses, while still using sophisticated methods to fill in incomplete information and correct some larger-scale errors*; 4D-Var can do this. It is not clear that the EnKF, with its limited ability to

* We cannot expect to have sufficient observations to define directly what we need to know, since observations are relatively expensive compared to data assimilation systems, so it is more cost effective to reduce the observations and have a sophisticated assimilation.

fit detailed observations without using a very large ensemble*, will suit this scenario as well as variational methods. It may be better to consider the hybrid methods described in section 5. These could use high-resolution 4D-Var, augmented with ensemble covariances, to model the evolution within each short† time window, and to extract tendency information from observation in the same window. A cheaper, lower-resolution, ensemble method evolves perturbations between cycles‡.

In order to run high-resolution NWP systems, most centres will continue to use limited-area models§. As discussed in subsection 4(g), boundary conditions from a global ensemble are needed when generating a limited-area ensemble; it cannot simply perpetuate itself using the EnKF methods of section 2. So the hybrid methods look particularly attractive for limited-area, mesoscale assimilation.

APPENDIX A

To show that a perfect observation removes a degree of freedom from the ensemble

By subtracting (4) from (6), and then substituting (8), it is easy to show that

$$\mathbf{x}_i^a - \bar{\mathbf{x}}^a = \sum_{j=1}^N (\mathbf{x}_j^f - \bar{\mathbf{x}}^f) w_{j,i} \quad \text{for } i = 1, N. \quad (\text{A.1})$$

If the analysis has just assimilated an error-free observation, all the analysed ensemble members, and the mean, will fit the observation. Applying the (linear) observation operator H to (A.1) gives

$$0 = \sum_{j=1}^N (H\mathbf{x}_j^f - H\bar{\mathbf{x}}^f) w_{j,i} \quad \text{for } i = 1, N. \quad (\text{A.2})$$

So, as long as the $H\mathbf{x}_j^f$ are different, the matrix of $w_{j,i}$ is degenerate, and adding the error-free observation will remove a degree of freedom from the ensemble.

APPENDIX B

Use of a Schur product in calculating x and y

Consider element (l, m) of the covariance of $\mathbf{x}$ given by

$$\mathbf{x} = \bar{\mathbf{x}}^f + \mathbf{X}^f \circ \boldsymbol{\alpha} \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}. \quad (\text{B.1})$$

* Snyder and Zhang (2003) successfully assimilated 2 km simulated Doppler radar observations in an EnKF with 50 members. But their identical-twin experiment was set up so there were no large-scale errors, so that they could use a covariance localization of 5 km without damaging the analysis of large scales.

† Lorenc (2003) discusses why 4D-Var can only assimilate information on scales which do not behave chaotically over the window. For small scales that restricts us to a short window.

‡ This is similar to the reduced rank Kalman filter developed by Mike Fisher at ECMWF (personal communication).

§ Others may avoid the boundary problem using a stretched global grid as in Côté *et al.* (1998).

We show that this is the same as element (l, m) of $\mathbf{P}_e^f \circ \mathbf{C}$. I use x'_i to denote the i th column of $\mathbf{X}^f$:

$$\begin{aligned}
 \langle (x(l) - \overline{x^f(l)})(x(m) - \overline{x^f(m)}) \rangle &= \left\langle \sum_{i=1}^N \alpha_i(l) x'_i(l) \sum_{j=1}^N \alpha_j(m) x'_j(m) \right\rangle \\
 &= \sum_{i=1}^N \sum_{j=1}^N x'_i(l) x'_j(m) \langle \alpha_i(l) \alpha_j(m) \rangle \\
 &= \sum_{i=1}^N \sum_{j=1}^N x'_i(l) x'_j(m) C(l, m) \delta_{ij} \quad (\text{B.2}) \\
 &= \sum_{i=1}^N x'_i(l) x'_i(m) C(l, m) \\
 &= P_e^f(l, m) C(l, m) \\
 &= (\mathbf{P}_e^f \circ \mathbf{C})_{l,m}.
 \end{aligned}$$

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